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Early Detection of Paddy Disease and Severity based Classification Using variants of CNN - A Step Towards Smart Agriculture

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***Abstract*—Agriculture is the backbone of India, and in the southern region, paddy is the main crop, grown twice a year. Over the years, paddy fields have been affected by different diseases, out of which three diseases are predominantly found in the Mysore, Mandya, and Srirangapatna regions, namely leaf blast, sheath blight, and brown spot. Early disease detection is crucial to prevent crop loss, but farmers often struggle to identify these diseases with the naked eye, making the process time-consuming and inaccurate. Manual inspection is not cost-effective, hence to address this issue, we have employed a Deep learning CNN using the InceptionV3 model for automated disease detection. Our dataset, consisting of 1,250 images from Zonal Agricultural Research Station (ZARS) , Mandya. The InceptionV3 model demonstrated high reliability in disease classification. Our model not only detects diseases but also estimates their severity, helping farmers determine whether the disease is curable or non-curable. This decision-support system enables early identification of diseases, improving disease management and reducing crop loss in infected fields.**

Keywords – Leaf Blast, Deep Convolution Neural Network, Disease Severity(1 to 9) Estimation, InceptionV3 ;

I. INTRODUCTION

India is one of the largest producers of rice, with over 46 million hectares of land dedicated to paddy cultivation, feeding millions of people. Despite rice being a staple food for over 60% of India's population, only a small percentage of farmers, around 2-3%, are actively involved in paddy cultivation. This highlights the need to protect paddy crops from diseases to ensure food security. Early disease identification is essential, as failing to assess the severity can lead to improper pesticide use, causing financial loss and potential crop damage. In this study, we collected data on major paddy diseases from the Zonal Agricultural Research Station (ZARS) [1] in Mandya. We studied three common diseases Brown Spot ,Sheath Blight, and Leaf Blast which significantly impact crop yield and quality. The odd scaling for disease severity was conducted under the guidance of Dr. V. B. Sanath Kumar, Professor and Head, Department of Plant Pathology, Mandya.

"Fig.1" shows Leaf Blast is caused by the fungus Magnaporthe oryzae. It begins as small, grayish-green lesions on leaves that expand into brown, diamond-shaped spots. In severe cases, the fungus spreads to stems and grains, leading to plant collapse. It significantly reduces photosynthesis, weakens the plant, and start spreading to other crops after reaching Scale 5. It can cause widespread crop loss if not treated early. Simillarly "Fig. 2" shows Sheath Blight caused by Rhizoctonia solani, initially appears as water-soaked lesions on the leaf sheath, which later turn grayish-white with brown margins. The disease spreads rapidly through dense plant canopies, weakening the structure and leading to plant lodging (bending over), and start spreading to other crops after reaching scale 5. Poor grain filling and reduced productivity are common effects, particularly in humid conditions. Brown Spot caused by Bipolaris oryzae is as shown in "Fig. 3" manifests as circular brown lesions with yellow halos on leaves and grains. The disease weakens the plant by disrupting photosynthesis, resulting in poor grain quality, yield loss and starts spreading to other crops after reaching Scale 5. Infected grains become discolored and unsuitable for consumption, further impacting farmers' profits.



Fig. 1.Paddy leaf affected by Leaf Blast disease scale 5

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Fig. 2. Paddy leaf effected by Sheath Blight scale 7



Fig. 3. Paddy leaf effected by Brown Spot scale 3

To effectively address paddy diseases, developed a CNN based model that not only identifies the disease but also evaluates its severity on an odd scale ranging from 1 to 9.A senior plant pathologist annotated each image, first identifying the disease class and then assigning a severity rating on a 1–9 ordinal scale based on visual yellowness and microscopic measurements of lesion radius. This helps farmers decide if the disease can be treated (severity <5) or if the crop is too damaged to recover (severity >=5). If the disease is treatable, they can use suitable pesticides to save their crops. If not, they can avoid spending money on ineffective treatments and spreading of diseases, leading to better resource use and improved farming productivity. We trained and tested our model using the InceptionV3 architecture, which proved to be the most suitable for identifying complex disease patterns.

II. Problem Statement

Disease severity plays a crucial role in effective crop management as improper pesticide use can further degrade soil health and impact on soil fertility for a longer duration of time. In order to address this problem there is a need for developing a systematic approach that not only detects diseases in timely manner but also determines the severity of the disease. By categorizing infected leaves into curable and non-curable stages farmers can take decisions about whether pesticide application is necessary or eradicating the effected plant to stop the further spreading of the diseases. A smart system using deep learning can help farmers detect plant diseases better, use minimal pesticide, and support sustainable farming.

III. Literature Survey

Jagamohan Padhi et al. [2] introduced a hybrid CNN model with simulated thermal imaging, converting 5,932 images to thermal form to capture early stress signals. Among 18 CNNs, Darknet53 performed best, and integration with an SVM classifier raised performance to 99.43% accuracy and an F1-score of 0.99, showing strong potential for real-time disease detection

Priyanka Kulkarni et al.[3] they have used VGG16 for classifying rice leaf diseases under varied lighting and background conditions. Model trained and tested for 1649 images of three prevalent diseases such as brown spot, rice leaf blast, and rice leaf blight. They used the SoftMax function for output and trained the model using the Adam optimizer, reaching 95% accuracy with VGG16. However, VGG16 isn't the best fit for agriculture tasks since leaf images often have complex backgrounds.

V. Sahasranamam et al. [4] proposed a paddy leaf disease detection system using DenseNet-121 on the Paddy Doctor dataset of size 10,407 images, including multiple disease and healthy classes. The model achieved 97.16% accuracy, outperforming ResNet-50 (94.81%) and VGG-16 (62.13%). Dense connectivity improved feature reuse and reduced over fitting, making the model effective even on smaller datasets.

Yogesh Kumar Rathore et al. [5] proposed a lightweight LW17 model for the detection of rice leaf diseases. They compared the LW17 model with pre-trained models such as Xception, VGG19, DenseNet201, InceptionV3, and MobileNet. The model was trained using 120 images of three disease classes: Bacterial Leaf Smut, Leaf Blight, and Brown Spot with 40 images in each category. The model obtained an accuracy of 93.75% and helps farmers protect their crops at an initial stage.

G. Indiravathi et al. [6] have used transfer learning with the VGG16 model to classify rice leaf diseases into leaf blight , brown spot and bacterial leaf blight. They trained their model on 1,509 leaf images and tested it on 647 different images, achieving 92.5% accuracy. The authors have suggested to the cross-validation process in future on real-time collected data to improve the outcomes.

Pandiyaraju V. et al. [7] developed SwiSENet, a channel attention–driven CNN framework extending SENet with Squeeze-and-Excitation blocks and Swish activation. Trained on 5,932 images, including bacterial blight, blast, brown spot, and tungro diseases, the model achieved 99.74% accuracy with high precision and recall.

Shankarnarayanan Nalini et al. [8] proposed a novel DNN-CSA model for paddy leaf disease detection. Model integrates a deep neural network with Crow Search Algorithm to model

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optimization . K-means clustering and thresholding were used to isolate diseased regions before feature extraction. The model achieved 96.96% accuracy, 95.92% precision, and 96.41% recall. When compared to a 10-fold validated SVM, DNN-CSA outperformed with over 9% higher scores. This model shows high accuracy with reduced computational workload for real time agricultural use.

Muhammad Juman Jhatial et al. [9] has proposed rice leaf disease detection using a deep learning model known as YOLOv5 and has led to increased crop productivity by detecting the plant diseases with higher accuracy at an earlier stage. They have used Google Colab platform to train, test, and validate over 400 images of infected leaves. They have investigated four well-known diseases frequently occurring in rice leaves such as bacterial blight, tungro, blast and brown spot. The experimental results show that they achieved 1.00 precision, 0.94 recall, and 0.62 mAP. However, their mAP score of only 0.62 is considered low.

Ghazanfar Latif et al. [10] proposed detection of rice plant diseases using VGG19-based transfer learning model. They used 5 different classes to train the model bacterial leaf blight,brown spot, leaf blast, narrow brown spot and leaf scald. With the help of VGG19 they achieved an high average accuracy of 96.01% when compare it to the GoogleNet and AlexNet.

Mustafizur Rahman Konik et al. [11] have investigated and detected paddy leaf diseases such as Leaf Blight, Brown Spot, and Leaf Blast using the k-Fold cross-validation technique for classification. They employed Logistic Regression, SVM , Decision Tree, , and Random Forest to automatically eliminate noise and reduce the time required to measure the impact of paddy leaf diseases, ensuring minimal manual intervention. The results clearly highlight that Random Forest provides a higher accuracy of 94.16% in contrast to the other techniques.

Mahir Tasfe et al. [12] conducted a systematic and structured survey to explore various deep learning methods for paddy leaf disease identification. Their study highlights the effective use of Vision Transformers, CNNs, and hybrid models for disease detection. They also addressed several research gaps, including the lack of large scale datasets, the absence of standard and accurate methods for disease severity detection, and the limited availability of segmentation data.

Drawbacks of Existing System

- Most current systems focus on identifying diseases but fail to determine their severity, which is crucial for effective treatment.
- Many models are trained on generic datasets, making them less useful for plant varieties specific to certain regions.
- Without severity classification, farmers may waste money on pesticides for non-curable crops, leading to financial losses and resource waste.
- Farmers must physically bring crop samples to Zonal Agriculture research Stations for inspection, which is time-consuming and inconvenient.

ADVANTAGES OF PROPOSED SYSTEM

- InceptionV3 is one of the CNN models used to produce accurate leaf disease classification along with disease identification.
- It also assesses the severity of the disease by determining whether it is curable or non-curable.
- Farmers need not bring their crops to Zonal Agriculture Research Stations for inspection, making disease detection faster and easier.
- This helps farmers determine the proper use of pesticides without overuse or misuse.

IV. SYSTEM DESIGN

A) System Architecture

Our system follows a structured process to detect paddy leaf diseases and assess their severity, as shown in “Fig. 4”. First, images of paddy leaves are collected, resized, and enhanced using data augmentation techniques to improve model accuracy. Images were augmented during training with rotation (±15°), horizontal flip (p=0.5), random scale (0.9–1.1), and shear (±10°), plus photometric jitter (brightness/ contrast /saturation 0.8–1.2) and mild Gaussian noise; no augmentation was applied to validation/test data. The model then extracts important features from these images using convolutional and pooling layers, succeeded by fully connected layers for deeper analysis. In the prediction stage, the model classifies the disease using the Softmax function and determines its severity using regression. Based on the severity score, the decision module categorizes the disease as curable (severity <5) or non-curable (severity >=5).

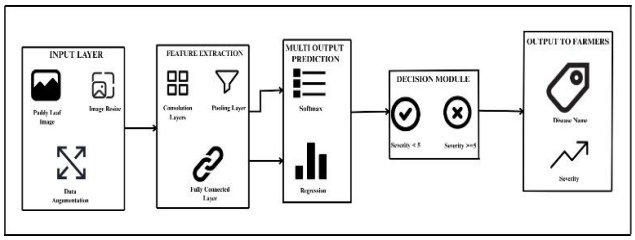


Fig. 4.: System Architecture diagram to detect disease and its severity

B) State Diagram

The state diagram, as shown in “Fig.5”, and the process begins when the user uploads an image of the paddy leaf. The system then moves to the Pre-Processing state, where the image is enhanced and prepared for further analysis. Next, the system transits to the Feature Extraction state, where essential characteristics such as texture, color and patterns are identified to help with classification. Patterns refer to the unique visual structures on the leaf, such as spots, streaks, or lesions, which implies the presence and severity of a disease.

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Once the features are extracted, the system enters the Disease Classification state, determining whether the leaf is affected by a disease or a leaf is healthy. At the same time, the Severity Analysis state checks how serious the infection is, using a odd scale of 1 to 9, to decide if the disease can be curable or non curable. After these analyses, the system enters the Display Results state, where it presents the identified disease, its severity score, and recommendations for the farmer.

The system transitions to the Decision State, where the user is prompted to either upload another image (looping back to the initial state) or exit the system if no further analysis is needed.

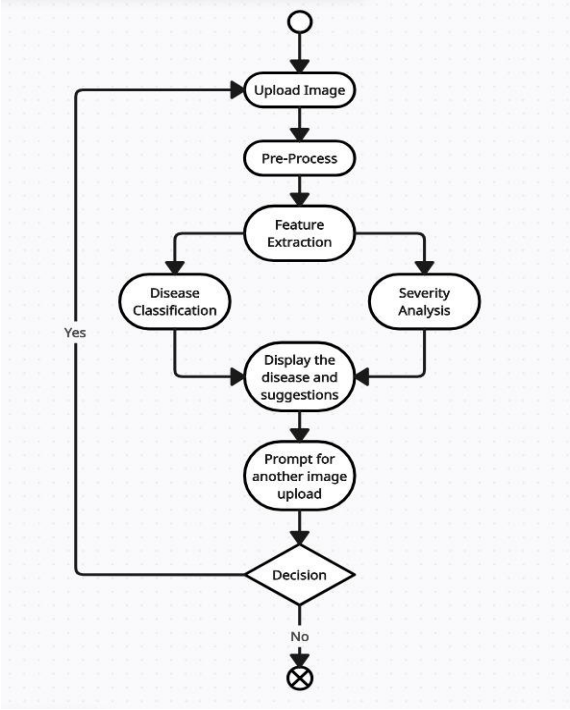


Fig. 5. State Diagram

V. System Implementation

a) Phases of implementation

Phase 1: Data Collection and Preparation

The dataset was collected from Zonal Agricultural Research Station, Mandya and organized into different categories based on disease type and severity level. Filtered the dataset in order to improve accuracy. Then, we resized, rescaled, and applied image augmentation (like rotation and flipping) to help the model learn better. The data was divided into 80% for training (1004 images) and 20% for testing (250 images) the model.

Phase 2: Model Selection and Training

We used pre-trained deep learning models ResNet50, EfficientNetB0, and InceptionV3 for disease classification and severity identification. ResNet50, with its deep residual connections, helps in capturing complex features effectively. EfficientNetB0, known for its optimized architecture, balances accuracy and efficiency. InceptionV3, with its multi-scale feature extraction, enhances the model's ability to detect diverse disease patterns. The model was trained to identify the disease and measure its severity at the same time. Categorical cross-entropy loss used for classification and mean squared error (MSE) for severity prediction. To improve performance, we used the Adam optimizer for better training. The model's performance was measured using accuracy (for classification) and mean absolute error (MAE) (for severity).

Phase 3: Data Processing for Training

Custom Data Generator Design involves building a custom data generator that can feed both classification and regression labels simultaneously into the model during training. This generator integrates with the image data pipeline and handles batch processing to ensure that the multi-task learning objectives are met without compromising the flow of data. It was an important component to align the tasks and process the dataset correctly for training.

Phase 4: Model Training and Evaluation

We trained our CNN models ResNet50, EfficientNetB0, and InceptionV3. The models were tested for accuracy in identifying diseases and in predicting severity. After comparing results, we found that InceptionV3 performed the best with the highest accuracy and the lowest MAE value used to track the accuracy during classification.

b) Core model implementation

The Inception-v3 model of this system is designed to perform multi-task learning, classify the paddy disease and predict its severity (scale from 1 to 9) . Inception-v3 is the deep learning model which having the ability to detect complex patterns in images. The pre trained model is first loaded with ImageNet weights, but its layers are frozen to retain its learned features. To make the model suitable for the task, a shared feature extractor is added, which further processes the image and extracts meaningful features before passing them to the classification and regression heads. The feature extractor consists of global average pooling, which reduces dimensionality, followed by a fully connected layer with ReLU activation and 128 neurons. The model's multi-task learning is then handled by two distinct heads. The four classes such as Healthy, Leaf Blast, Sheath Blight, and Brown Spot are used by the classification head to determine the type of disease. It guarantees that the output is a probability distribution by using a softmax activation function. The regression head can handle continuous values since it uses a linear activation function to forecast the disease's severity scale (1–9). Severity was estimated via a regression head trained on expert-assigned 1–9 ordinal scores, using shared features with the classifier; outputs were mapped to the 1–9 range and assessed with MAE and quadratic weighted kappa. Ten epochs of training enable the model to pick up significant patterns. After which, the model is downloaded and saved as a file allowing predictions on fresh photos of paddy leaves to identify the illness kind and severity.

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Learning Curve: Training Accuracy Trends Over Epochs

"Fig. 7" illustrates the learning curve of the InceptionV3 model, showing the training accuracy progress over the epochs. The curve shows that the model gets better as it trains and becomes more stable in the later epochs, meaning it has learned well.

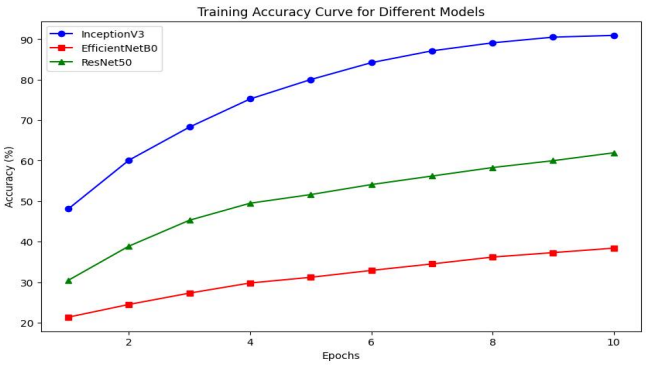


Fig. 7. Training Accuracy over Epoch curve

VI. TESTING

Table 1 Test Case Table of Disease classification and severity prediction

Test case ID	Sample Input	Expected Disease output	Expected Curability Output	Obtained Disease Output	Obtained Severity Output	Status (Pass/Fail)
TCU01	Paddy Leaf image with leaf blast (scale 5)	Leaf Blast	Curable	Leaf Blast	Scale 5	Pass
TCU02	Paddy Leaf image with Sheath Blight (scale 3)	Sheath Blight	Curable	Sheath Blight	Scale 3	Pass
TCU03	Paddy Leaf image with Brown Spot (scale 9)	Brown Spot	Not Curable	Brown Spot	Scale 9	Pass
TCU04	Healthy Paddy leaf image	Healthy	-	Healthy	-	Pass
TCU05	Paddy Leaf image with Sheath Blight (scale 7)	Sheath Blight	Not Curable	Sheath Blight	Scale 7	Pass
TCU06	Paddy leaf image with mixed diseases (Leaf Blast + Brown Spot)	Leaf Blast	-	Either of the disease	-	Fail
TCU7	Paddy leaf image with unseen disease (not in training data)	Error / Unknown Disease	-	Incorrect classification	-	Fail

In order to simulate real world agricultural scenarios, observing factors such as several disease types, varying severity levels, multiple infections within a single leaf, and low-quality images which consist of background noise were taken into consideration. By including such various conditions, the testing phase guarantees that the model is robust enough to handle complex practical field applications where image quality and disease symptoms can vary significantly. Each test case consists of a Test Case ID, Sample Input, Expected Disease Output, Expected Severity, Obtained Disease Output, Obtained Severity, and Status with Pass or Fail as it shown in Table 1. The pass or fail status is decided by comparing the expected disease classification and severity prediction with the model's actual output, assuring consistency and accuracy in disease identification. The test performed very well in the most of the cases but the challenge raised when an image contained multiple overlapping diseases, indicating the need for further improvements in handling mixed infection.

VII. Result Analysis

Three different models of CNN namely ResNet50, EfficientNetB0 and InceptionV3 where considered after through literature survey for implementation and analysis of results by taking an overall dataset consisting of 1,250 images. The graph in "Fig. 10." Clearly indicates that InceptionV3 model outperforms the other two models w.r.t F1-score, precision and recall as it uses a combination of factorized convolutions, auxiliary classifiers, and efficient feature extraction. InceptionV3 architecture has an advantage over the traditional CNN architectures as it can represent complex patterns due to the multi-scale effective feature extraction ensuring that diagnosis of even the smallest symptoms of the disease is assured.ResNet50 has exhibited a moderate performance as a result of mis classification in some cases with , precision = 0.4281, recall = 0.6200, and F1 score = 0.5017. Comparatively EfficientNetB0 has performed very poorly with precision = 0.1866 and F1-score = 0.2606, due to the fact that it uses lightweight architecture unable to learn a great deal about disease classification. Thus the InceptionV3 model clearly outperforms in disease classification with an F1-score of 0.9721, a precision of 0.9728, and a recall of 0.9720. When compare to ResNet50 and EfficientNetB0 using the same dataset

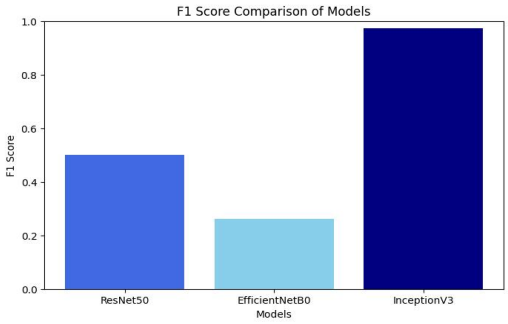


Fig. 10. Comparison of F1- Scores

The confusion matrices presents a vast understanding of each of the model's performance as shown in "Fig. 11." below. Struggle with fine-grained disease variations ResNet50 shows significant misclassification, particularly between leaf blast and sheath blight, reducing its overall effectiveness. Because of EfficientNetB0's reduced feature map sizes and sensitivity to training hyper parameters, leading to poor generalization it

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suffers from serious misclassification problems, almost all of its predictions concentrated into one class, suggesting its ineptness in differentiating several disease classes. InceptionV3, however, showed much higher performance than EfficientNetB0 and has been observed to misclassify few annotations, as there are clear diagonal trends in its confusion matrix. These results suggest InceptionV3 is certainly the best candidate for paddy disease detection and is, therefore, the ideal model to opt for practical applications.

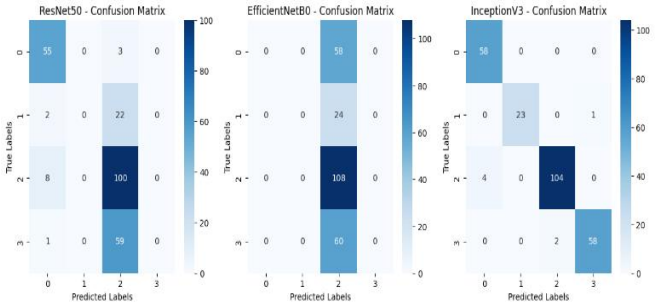


Fig. 11. Comparison of Confusion matrix

Further, the experimentation with different ratios of training and validation data to evaluate the model's performance was carried by using an 80:20 ratio (80% training and 20% validation), the model obtained its highest accuracy of 97.20%. As we increased the proportion of validation data, the accuracy slightly decreased. For the 70:30 split, the model achieved an accuracy of 93.87%, and with a 60:40 ratio, the accuracy was 93.60%. These results indicate that while increasing the amount of validation data slightly reduces the model's accuracy, it remains relatively consistent, demonstrating strong generalization across different splits.

VIII. CONCLUSION AND FUTURE ENHANCEMENT

This study addresses a key challenge in early detection and severity estimation of rice leaf disease, which is very crucial to prevent yield loss and maintain crop quality. To tackle this, an Deep Learning based model InceptionV3 was used achieving 97.21% accuracy. The system eliminates manual inspection at Zonal Agriculture Research Stations, making disease identification faster, more accurate, and cost-effective. The system also estimates disease severity on a scale of 1 to 9, allowing farmers to decide if the disease is curable or non curable, leading to better pesticide use and resource management. If widely adopted, this model can enhance farming practices, increase crop yield, and improve food security.

The future directions focus on addressing key challenges in rice leaf disease classification and enhancing its capability for recognizing multiple disease classes that can handle geographical variations in disease patterns, ensuring robust performance across different areas. Furthermore, integrating the model with IoT devices could allow for automated disease monitoring, providing real-time solutions for farmers. Additionally, the same model architecture can also be adapted to detect diseases in other crops, such as chili and tomato, expanding its application to a wider range of agricultural challenges.Additionally, the same model architecture can also be adapted to detect diseases in other crops, such as chili and tomato, expanding its application to a wider range of agricultural challenges.

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